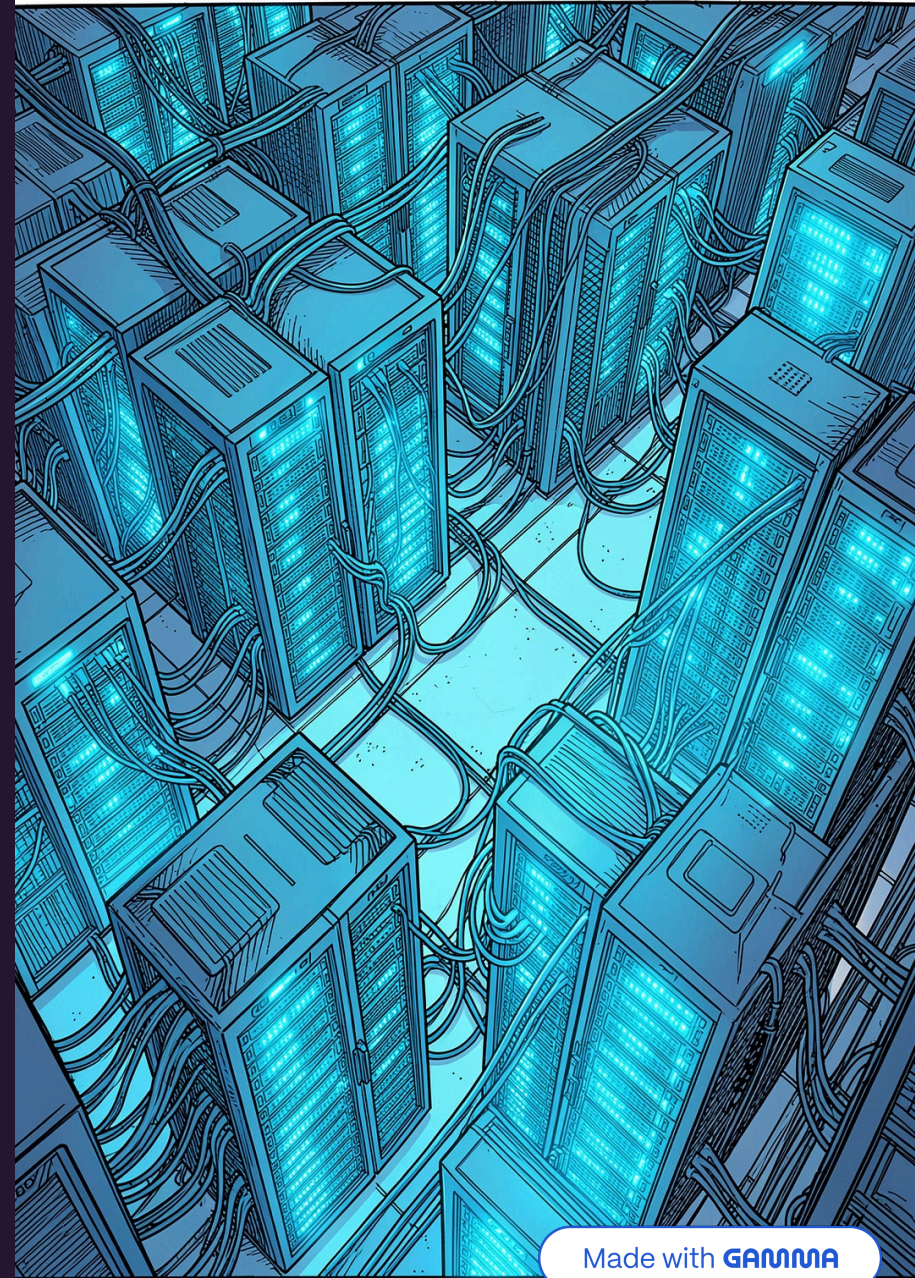


Rentlora Cloud Architecture

A modern property and booking platform built for enterprise-grade scalability, security, and performance on AWS.



Platform Overview

Rentlora is built for high availability, elastic scale, and secure service separation across AWS.

Microservices

Independent FastAPI services that scale and deploy separately.

Multi-AZ

Distributed across Availability Zones for resilience.

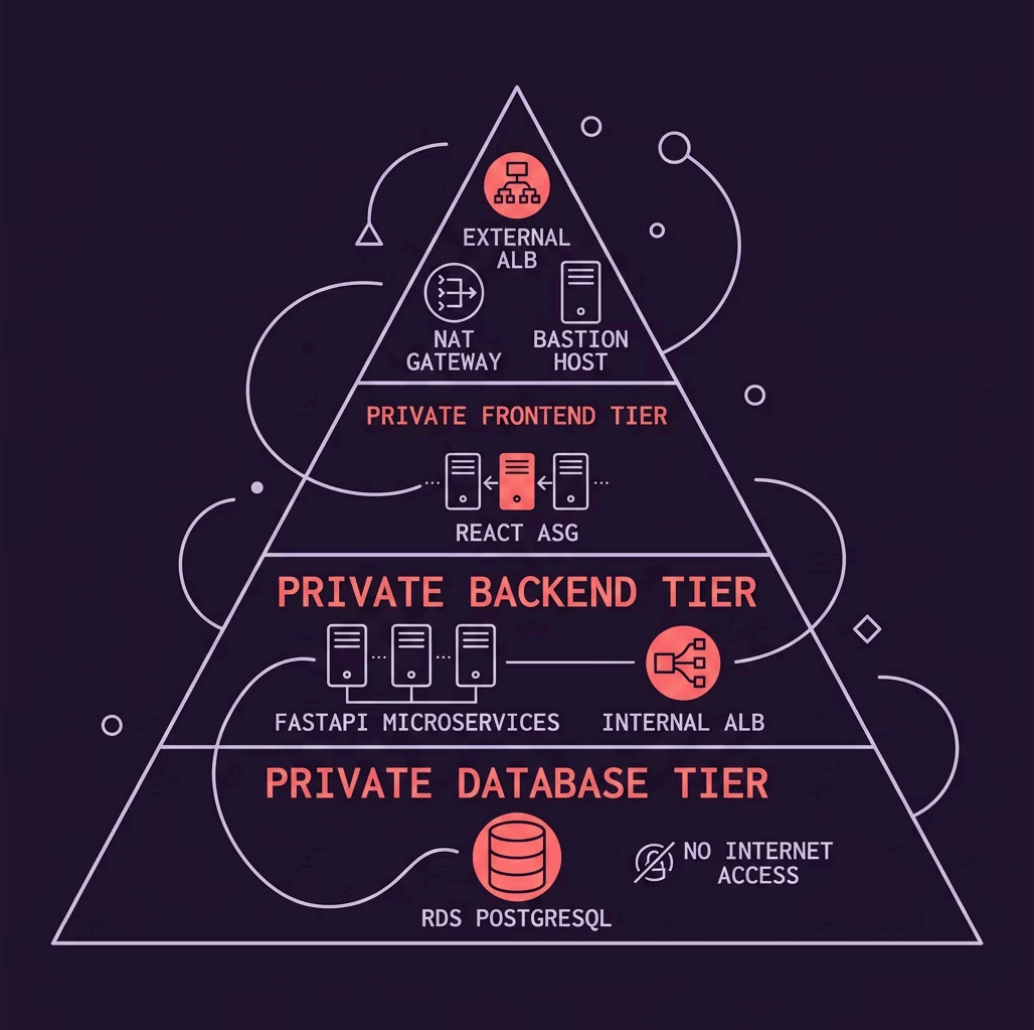
VPC Isolation

Tiered network controls with strict access boundaries.

Network Security & VPC Isolation

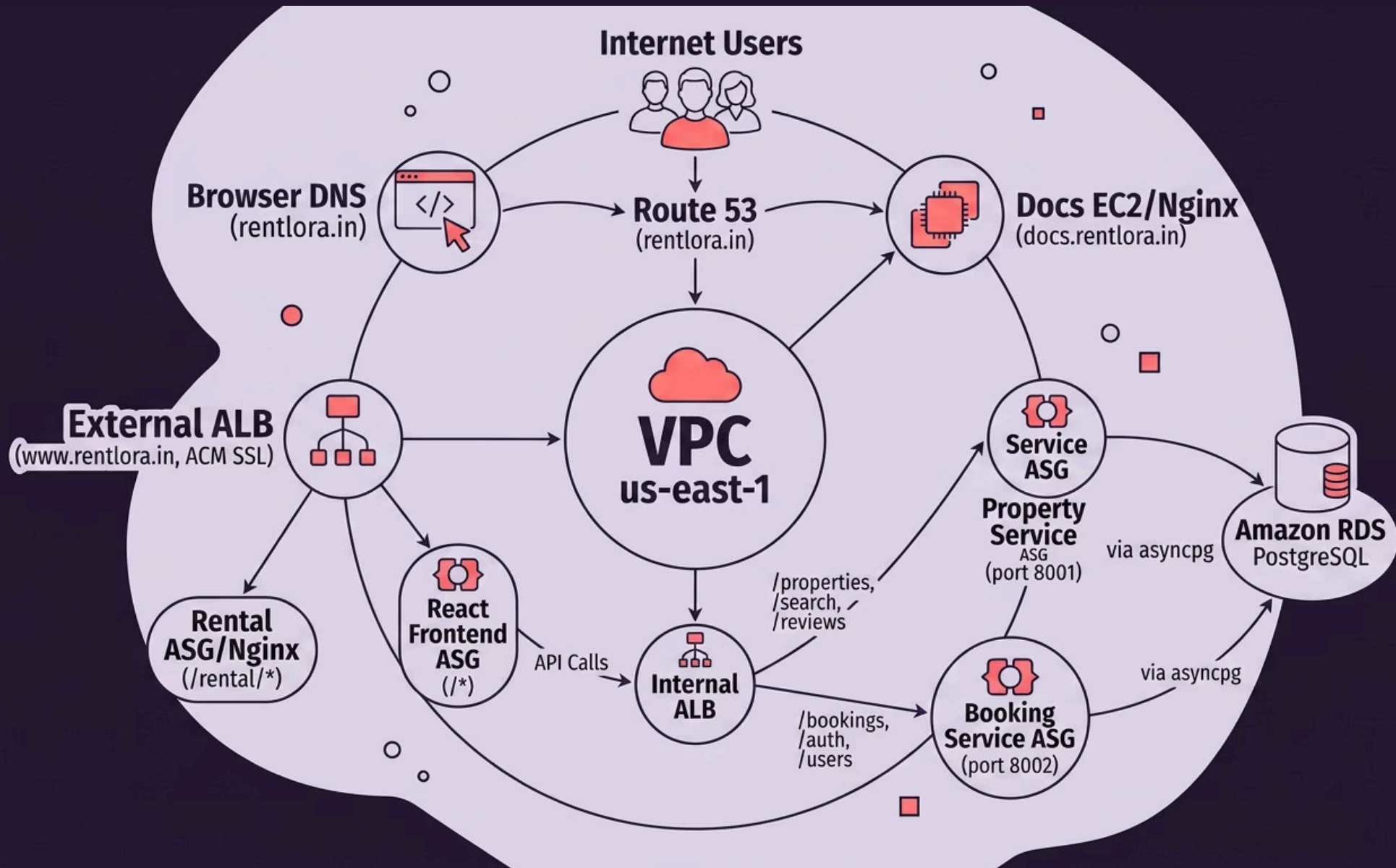
A custom multi-tier VPC in **us-east-1** enforces strict network segmentation across four subnet tiers.

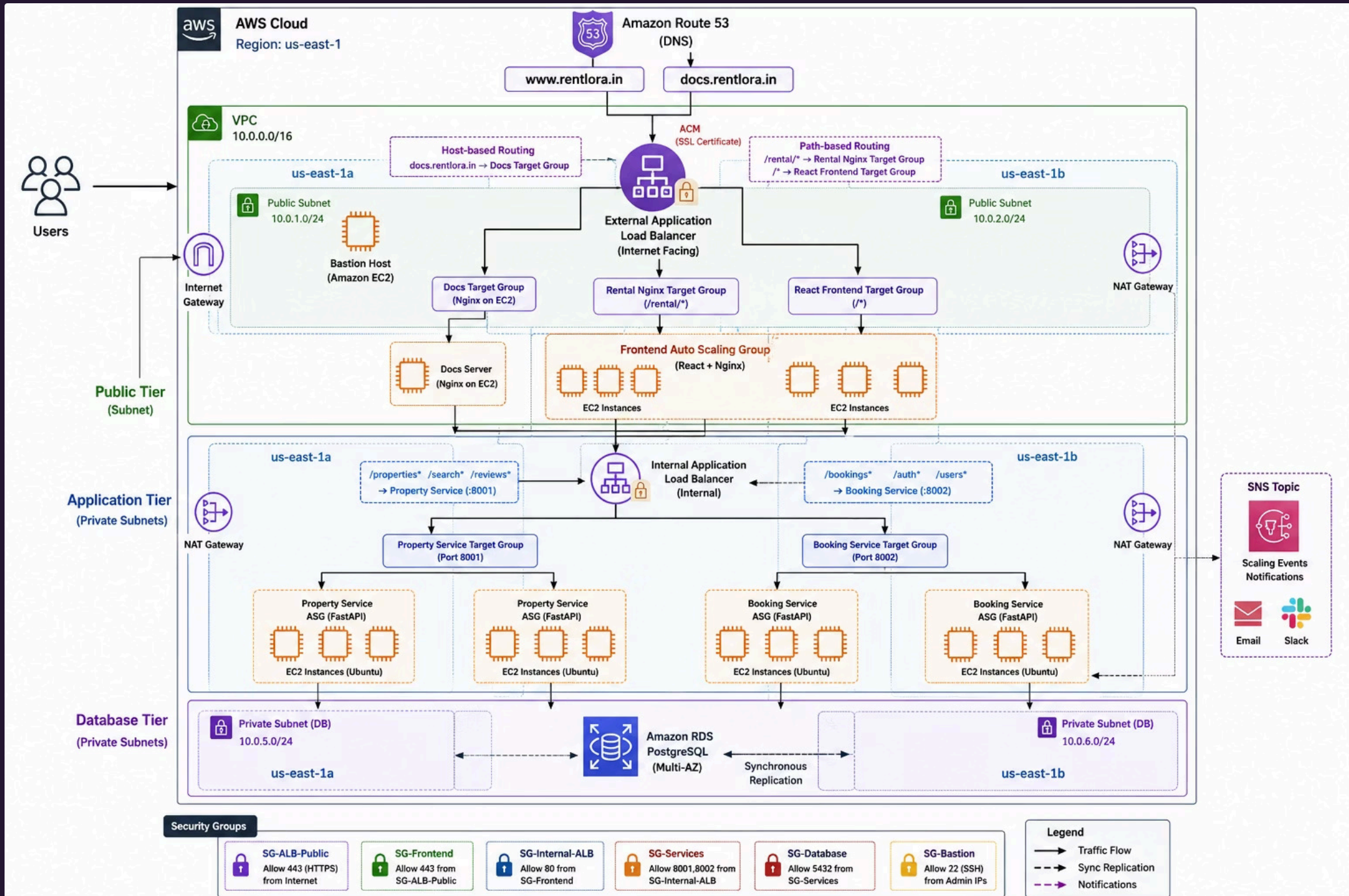
Public Subnet External ALB, NAT Gateway, Bastion Host.	Private Frontend Tier React ASG with no direct internet exposure.
Private Backend Tier FastAPI services behind Internal ALB.	Private Database Tier RDS PostgreSQL with no public access.



System Architecture

End-to-end request flow from internet users through DNS, ALBs, microservices, and the database layer.





Advanced Traffic Routing – External ALB

The **External Application Load Balancer** serves as the platform's single public entry point and routes traffic to the correct service.



Host-Based Routing

`docs.rentlora.in` is routed to the Docs EC2/Nginx server.



Path-Based Routing

`/rental/*` routes to the Nginx marketing cluster. `/*` routes to the React SPA frontend cluster.

Internal Service Routing – Internal ALB

The **Internal ALB** acts as a private API gateway, routing frontend requests to backend microservices without exposing them publicly.

Property Service ASG :8001

- /properties
- /search
- /reviews

Booking Service ASG :8002

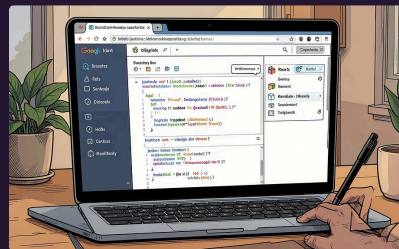
- /bookings
- /auth
- /users

Compute & Persistence Layer



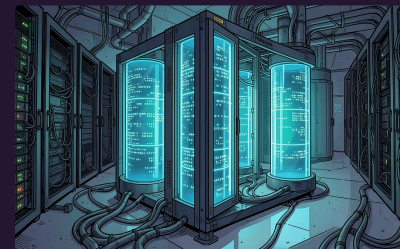
Compute Layer

Python 3.12 FastAPI microservices run on Miniconda-backed EC2 Auto Scaling Groups with Nginx for elasticity.



Frontend Layer

React SPA is served by an Auto Scaling Group behind the External ALB.

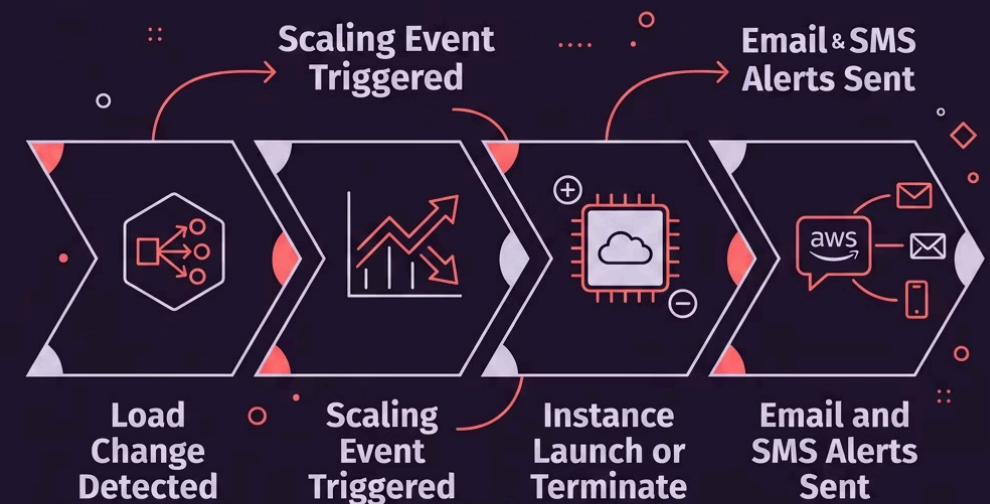


Database Layer

Amazon RDS PostgreSQL stays in private subnets with async access via `asyncpg`.

Auto Scaling Alerts & SNS Notifications

AWS SNS is integrated with Auto Scaling Groups to deliver **real-time notifications** on every scaling event — keeping the DevOps team immediately aware of infrastructure changes.



→ EC2 Launch Events

Alerts triggered when new instances spin up during scale-out.

→ EC2 Termination Events

Notifications on instance termination during scale-in.

→ Scale-Out / Scale-In Operations

Full visibility into capacity changes across all ASGs.

THANK YOU